reflection of the fact that what we see is not what there really is. Under the right circumstances we see particles, but our best current theories say that fields are more fundamental.

Gravity is the one part of physics that doesn't fit comfortably into the quantum-field-theory paradigm. You will often hear that "we don't have a quantum theory of gravity," but that's a bit too strong. We have an extremely good classical theory of gravity: Einstein's general relativity, which describes the curvature of spacetime. General relativity is itself a field theory—it describes a field pervading all of space, in this case the gravitational field. And we have very well understood procedures for taking a classical field theory and quantizing it, yielding a quantum field theory. Apply those procedures to the known fields of fundamental physics, and we end up with something called the *Core Theory*. The Core Theory accurately describes not only particle physics but also gravity, as long as the strength of the gravitational field doesn't grow too large. It is sufficient to describe every phenomenon that happens in your everyday experience, and quite a bit beyond—tables and chairs, amoebas and kittens, planets and stars.

The problem is that the Core Theory doesn't cover a number of situations beyond the everyday, including places where gravity becomes extreme, like black holes and the Big Bang. In other words, we have a theory of quantum gravity that is adequate when gravity is fairly weak, one that is perfectly capable of describing why apples fall from trees or how the moon orbits the Earth. But it's limited; once gravity becomes very strong, or we try to push our calculations too far, our theoretical apparatus fails us. As far as we can tell, this situation is unique to gravity. For all the other particles and forces, quantum field theories seem to be able to handle any situation we can imagine.

Faced with the difficulty of quantizing general relativity as we would any other field theory, there are a number of strategies that we might try. One is simply to think harder; maybe there is a good way to directly quantize general relativity, but it involves new techniques that we haven't needed for other field theories. A different approach is to imagine that general relativity isn't the right theory to quantize; maybe we should start with a distinct classical precursor, such as string theory, and then quantize that, hoping to build a quantum theory that includes gravity along with everything else. Physicists have been trying both of these approaches for